Tahanan Hub rev A
Home Assistant hub: CM5, NVMe, GbE with 802.3bt PoE, USB-C PD, Zigbee, Thread, Z-Wave socket, voice audio.
All inter-sheet connections are global labels.

Power input: USB-C PD sink, input OR-ing

File: power_in.kicad_sch

Regulators and sequencing

File: power_rails.kicad_sch

Compute Module 5 and system control

File: cm5.kicad_sch

NVMe (M.2 Key M) and USB 3.0 host ports

File: storage_usb.kicad_sch

Gigabit Ethernet and 802.3bt PoE powered device

File: ethernet_poe.kicad_sch

Zigbee, Thread and Z-Wave

File: radios.kicad_sch

Audio codec, microphone and speaker output

File: audio.kicad_sch

Tahanan

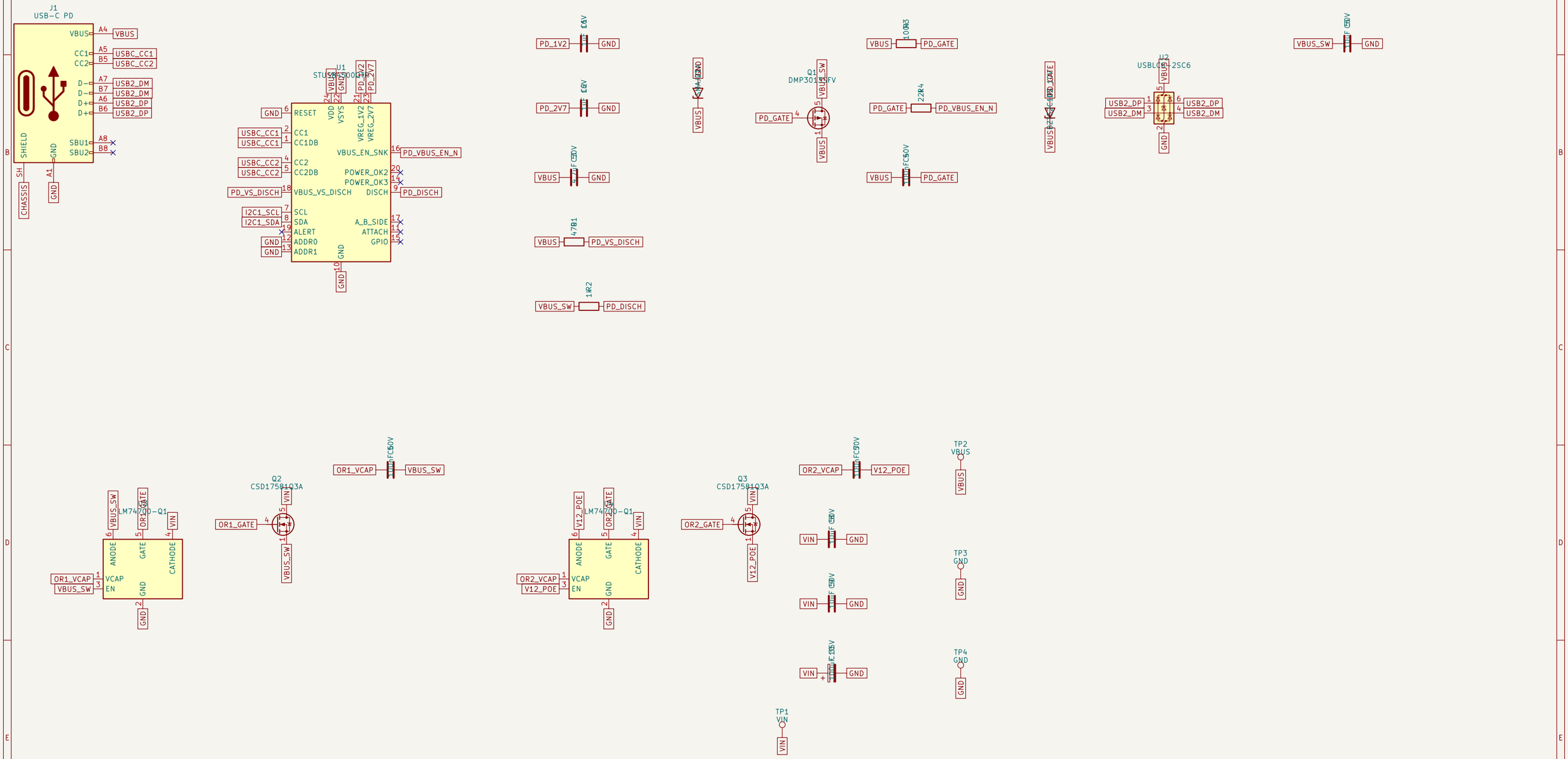
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Title: Tahanan Hub – top level

Size: A3	Date:	**Rev: A**
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Power input: USB-C PD sink, input OR-ing

USB-C PD sink: STUSB4500 autonomous controller. NVM must be programmed at factory test (PDO2 15V/3A, PDO3 20V/2.25A, POWER_ONLY_ABOVE_5V=1), see firmware/stusb4500. Q1 passes VBUS only after an explicit contract above 5V.
OR-ing: two LM74700 ideal-diode controllers. The higher of VBUS_SW (15-20V) and V12_POE (12V) feeds VIN, so USB-C takes priority automatically and neither source can back-feed the other. System UVLO is 8V (EN divider on U5).



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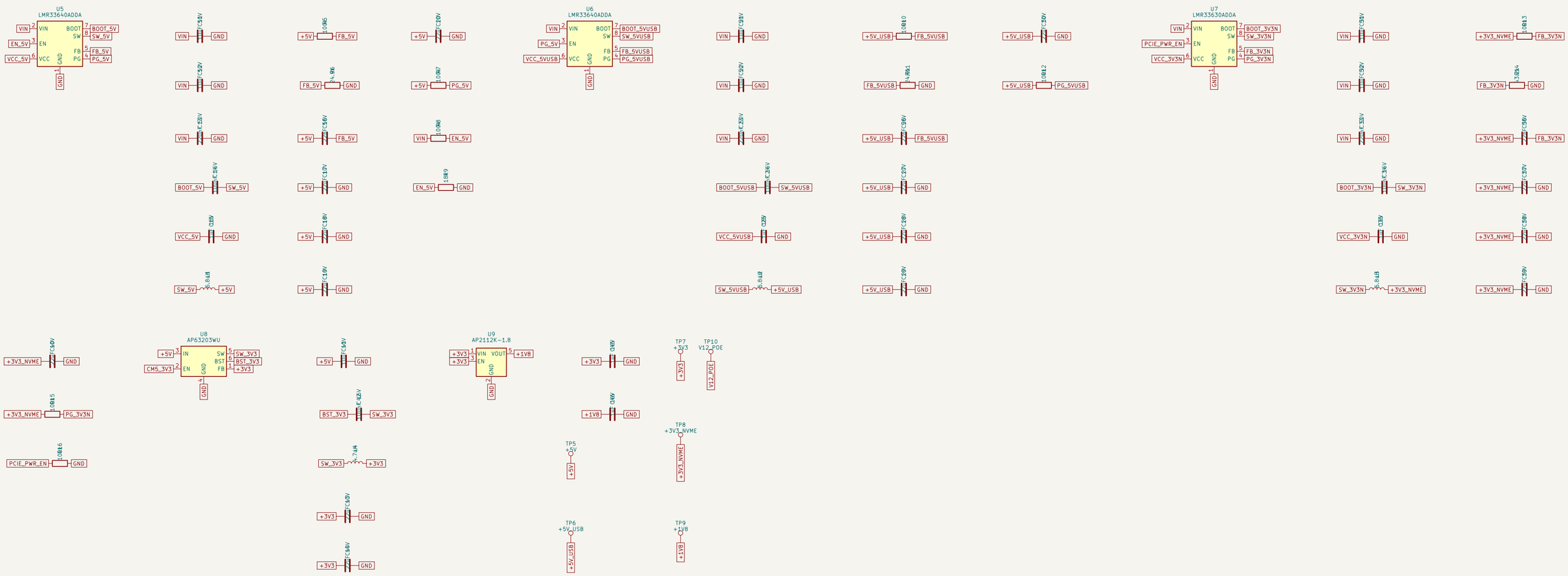
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File: power_in.kicad_sch

Title: Tahanan Hub – Power input: USB-C PD sink, input OR-ing

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Regulators and sequencing

Sequence: VIN>8V -> +5V (U5) -> PG_5V enables +5V_USB (U6). CM5 starts from +5V: CM5_3V3 then enables +3V3 (U8).
+3V3_NVME (U7) is enabled by PCIE_PWR_EN from the CM5. Nothing drives a CM5 pin before +5V is valid (CM5 datasheet 3.1).
LMR336x0: VFB=1.0V. 5V: 100k/24.9k. 3.3V: 100k/43.2k. fsw=400kHz, L=6.8uH, CIN 2x10uF+220nF (datasheet table 9-1).



PWR_FLAG +3V3 · PWR_FLAG +3V3A · PWR_FLAG +3V3_NVME · PWR_FLAG +5V · PWR_FLAG +5V_SPK · PWR_FLAG +5V_USB · PWR_FLAG CM5_3V3 · PWR_FLAG GND · PWR_FLAG MIC_VDD · PWR_FLAG POE_VIN_N · PWR_FLAG POE_VIN_P · PWR_FLAG VBAT · PWR_FLAG VBUS · PWR_FLAG VBUS_SW · PWR_FLAG VIN

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Sheet: /Regulators and sequencing/
File: power_rails.kicad_sch

Title: Tahanan Hub — Regulators and sequencing

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Compute Module 5 and system control

J3 carries module pins 1–100, J13 pins 101–200 (pad n = module pin 100+n). Stacking height 3.0mm.
CC1/CC2 are not used: PD is handled by U1, so set PSU_MAX_CURRENT=5000 in the bootloader EEPROM config.
GPIO map: 2/3 I2C1, 4–7 UART2 Zigbee, 8–11 UART3 Thread, 12/13 UART4 console, 14/15 UART0 Z-Wave socket,
16 codec reset, 17 user button, 18–21 I2S, 22/23 Zigbee RST/BOOT, 24/25 Thread RST/BOOT, 26 user LED, 27 mic-enabled sense.

J3
CM5 pins 1–100

1	GND	GND	2
3	Ethernet_Pair3_P	Ethernet_Pair1_P	4
5	Ethernet_Pair3_N	Ethernet_Pair1_N	6
7	GND	GND	8
9	Ethernet_Pair2_N	Ethernet_Pair0_N	10
11	Ethernet_Pair2_P	Ethernet_Pair0_P	12
13	GND	GND	14
15	Ethernet_nLED3	Fan_Tacho	16
17	Ethernet_nLED2	Ethernet_SYNC_OUT	18
19	Fan_PWM	EEPROM_nWP	20
21	LED_nACT	GND	22
23	GND	GPIO26	24
25	GPIO21	GPIO19	26
27	GPIO20	GPIO13	28
29	GPIO16	GPIO6	30
31	GPIO12	GND	32
33	GND	GPIO5	34
35	ID_SC	ID_SD	36
37	GPIO7	GPIO11	38
39	GPIO8	GPIO9	40
41	GPIO25	GND	42
43	GND	GPIO10	44
45	GPIO24	GPIO22	46
47	GPIO23	GPIO27	48
49	GPIO18	GPIO17	50
51	GPIO15	GND	52
53	GND	GPIO4	54
55	GPIO14	GPIO3	56
57	SD_CLK	GPIO2	58
59	GND	GND	60
61	SD_DAT3	SD_CMD	62
63	SD_DAT0	SD_DAT5	64
65	GND	GND	66
67	SD_DAT1	SD_DAT4	68
69	SD_DAT2	SD_DAT7	70
71	GND	SD_DAT6	72
73	SD_VDD_OVERRIDE	GND	74
75	SD_PWR_ON	VBAT	76
77	5V	GPIO_VREF	78
79	5V	SCL0	80
81	5V	SDA0	82
83	5V	CM5_3.3V	84
85	5V	CM5_3.3V	86
87	5V	CM5_1.8V	88
89	WL_nDisable	CM5_1.8V	90
91	BT_nDisable	PWR_Button	92
93	nRPIBOOT	CC1	94
95	LED_nPWR	CC2	96
97	CAM_GPIO0	GND	98
99	PMIC_Enable	CAM_GPIO1	100

J13
CM5 pins 101–200

101	USB_OTG_ID	PCIe_CLK_nREQ	102
103	USB_N	PCIE_nWAKE	104
105	USB_P	PCIE_PWR_EN	106
107	GND	GND	108
109	PCIe_nRST	PCIe_CLK_P	110
111	VBUS_EN	PCIe_CLK_N	112
113	GND	GND	114
115	MIPI0_D0_N	PCIe_RX_P	116
117	MIPI0_D0_P	PCIe_RX_N	118
119	GND	GND	120
121	MIPI0_D1_N	PCIe_TX_P	122
123	MIPI0_D1_P	PCIe_TX_N	124
125	GND	GND	126
127	MIPI0_C_N	USB3-0-RX_N	128
129	MIPI0_C_P	USB3-0-RX_P	130
131	GND	GND	132
133	MIPI0_D2_N	USB3-0-DP	134
135	MIPI0_D2_P	USB3-0-DM	136
137	GND	GND	138
139	MIPI0_D3_N	USB3-0-TX_N	140
141	MIPI0_D3_P	USB3-0-TX_P	142
143	HDMI1_HOTPLUG	GND	144
145	HDMI1_SDA	HDMI1_TX2_P	146
147	HDMI1_SCL	HDMI1_TX2_N	148
149	HDMI1_CEC	GND	150
151	HDMI0_CEC	HDMI1_TX1_P	152
153	HDMI0_HOTPLUG	HDMI1_TX1_N	154
155	GND	GND	156
157	USB3_1-RX_N	HDMI1_TX0_P	158
159	USB3_1-RX_P	HDMI1_TX0_N	160
161	GND	GND	162
163	USB3_1-DP	HDMI1_CLK_P	164
165	USB3_1-DM	HDMI1_CLK_N	166
167	GND	GND	168
169	USB3_1-TX_N	HDMI0_TX2_P	170
171	USB3_1-TX_P	HDMI0_TX2_N	172
173	GND	GND	174
175	MIPI1_D0_N	HDMI0_TX1_P	176
177	MIPI1_D0_P	HDMI0_TX1_N	178
179	GND	GND	180
181	MIPI1_D1_N	HDMI0_TX0_P	182
183	MIPI1_D1_P	HDMI0_TX0_N	184
185	GND	GND	186
187	MIPI1_C_N	HDMI0_CLK_P	188
189	MIPI1_C_P	HDMI0_CLK_N	190
191	GND	GND	192
193	MIPI1_D2_N	MIPI1_D3_N	194
195	MIPI1_D2_P	MIPI1_D3_P	196
197	GND	GND	198
199	HDMI0_SDA	HDMI0_SCL	200

J6
FAN

1	+5V
2	FAN_PWM
3	GND
4	FAN_TACH

J7
CONSOLE 3V3

1	GND
2	DBG_TXD
3	DBG_RXD

TP11 EEPROM_nWP
TP12 CM5_3V3
H1 M2.5 standoff 3mm
H2 M2.5 standoff 3mm
H3 M2.5 standoff 3mm
H4 M2.5 standoff 3mm
H5 M2.5 standoff 3mm
H6 M2.5 standoff 3mm
H7 M2.5 standoff 3mm
H8 M2.5 standoff 3mm

1R21 USER_LED LEDU_A
1R20 +3V3 LEDA_A
1R19 +3V3 LEDP_A
BT1 VBAT GND
4.7K +3V3 I2C1_SDA
4.7K +3V3 I2C1_SCL
100nF +5V GND
100nF +5V GND
100nF +5V GND
100nF +5V GND
100nF CM5_3V3 GND
100nF CM5_3V3 GND
RPIBOOT_N
PWR_BTN_N
USER_BTN_N
LED_PWR_N
LED_ACT_N

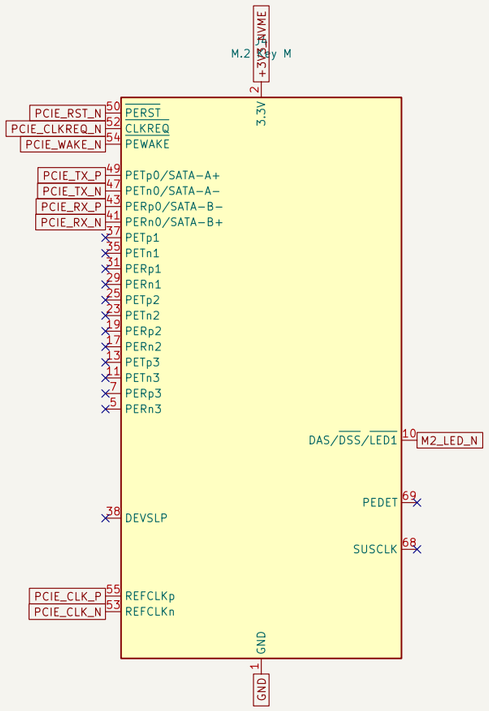
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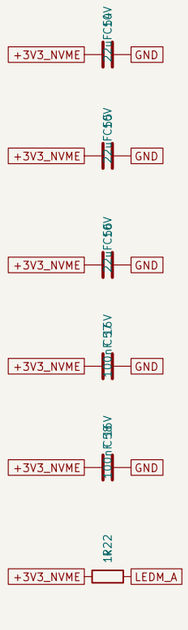
Title: Tahanan Hub – Compute Module 5 and system control

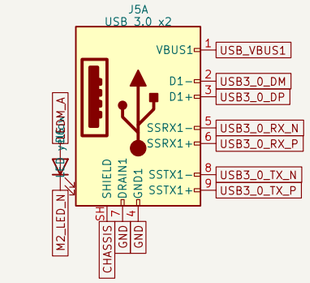
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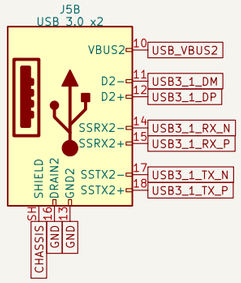
NVMe (M.2 Key M) and USB 3.0 host ports

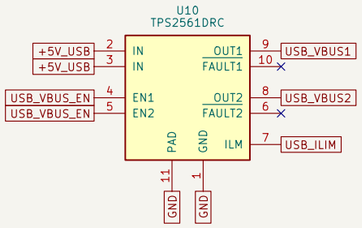
PCIe x1: TX AC-coupling capacitors are on the CM5, RX capacitors are on the SSD. Route as 90 ohm pairs (CM5 datasheet 2.3).
USB VBUS: TPS2561 dual switch, about 1.2A per port (R_ILIM 47k). Each port has 150uF bulk as USB requires 120uF minimum.

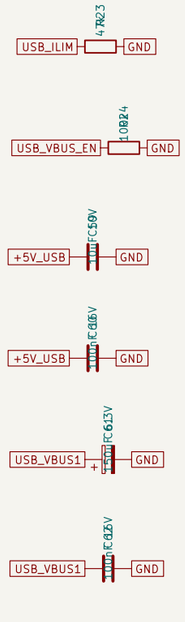


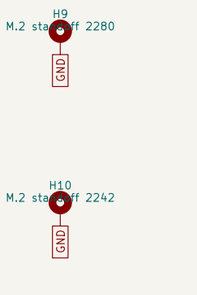


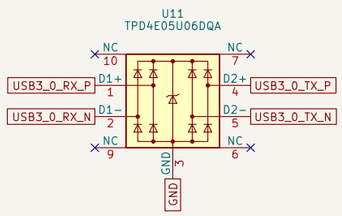


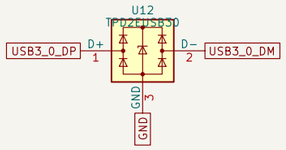


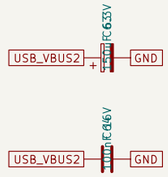


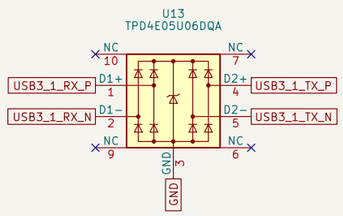


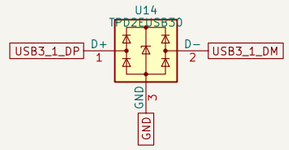












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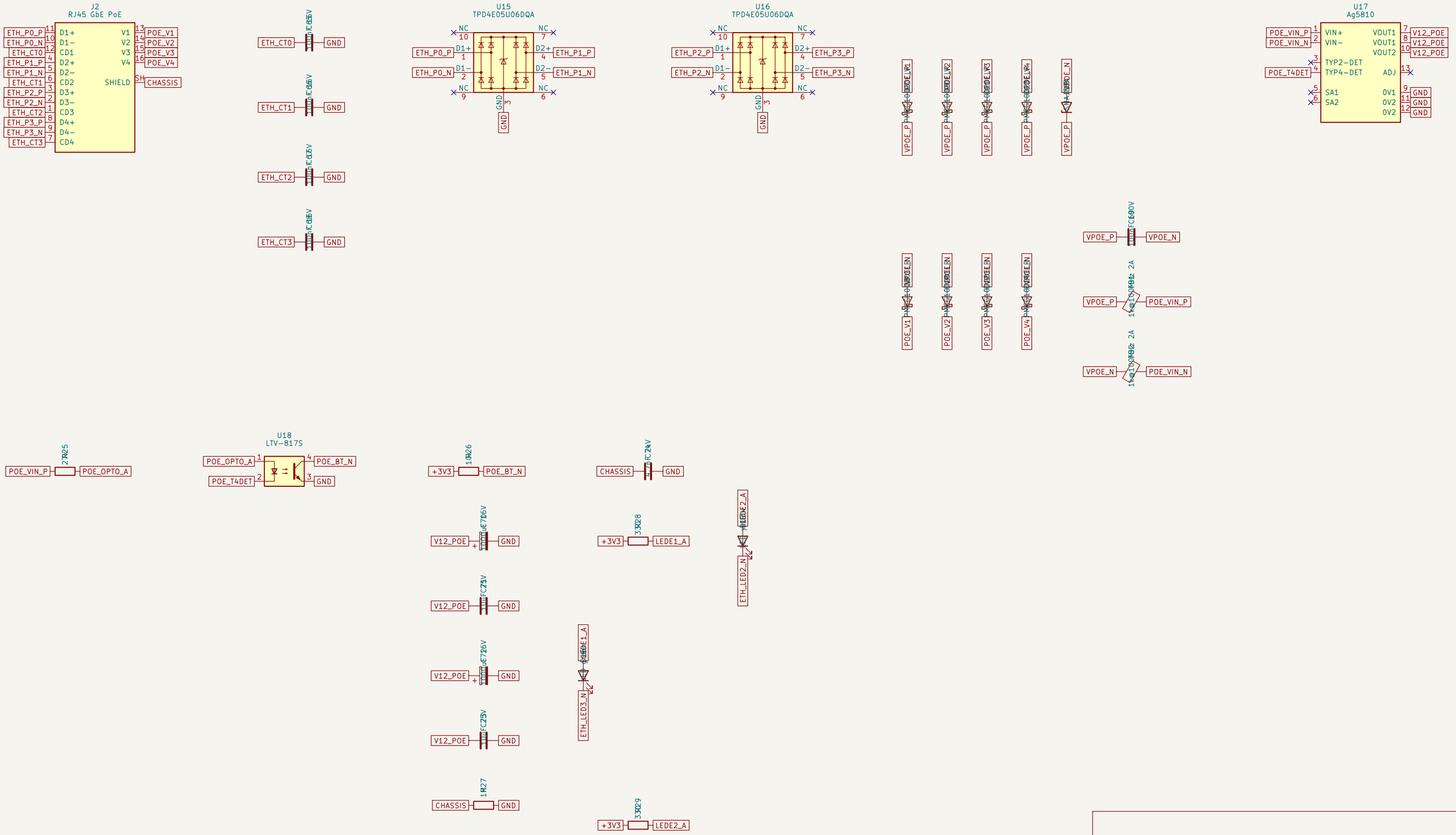
Sheet: /NVMe (M.2 Key M) and USB 3.0 host ports/
File: storage_usb.kicad_sch

Title: Tahanan Hub — NVMe (M.2 Key M) and USB 3.0 host ports

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Gigabit Ethernet and 802.3bt PoE powered device

ISOLATION: every net on the cable side of U17 (V1–V4, VPOE_*, POE_VIN_*, POE_T4DET, POE_OPTO_A) is the PoE primary domain.
Keep 2.5mm creepage to all other copper and no planes under it (net class PoE_Primary). Rated 1500V impulse (Ag5810 datasheet 4).
Two Schottky bridges (eight PMEG10020) accept power on either or both pairsets and either polarity. U17 presents Class 7; the
jack is rated 600mA per centre tap, which bounds PoE input at 60W PSE power. Opto U18 tells Linux when a Type 4 PSE is present.



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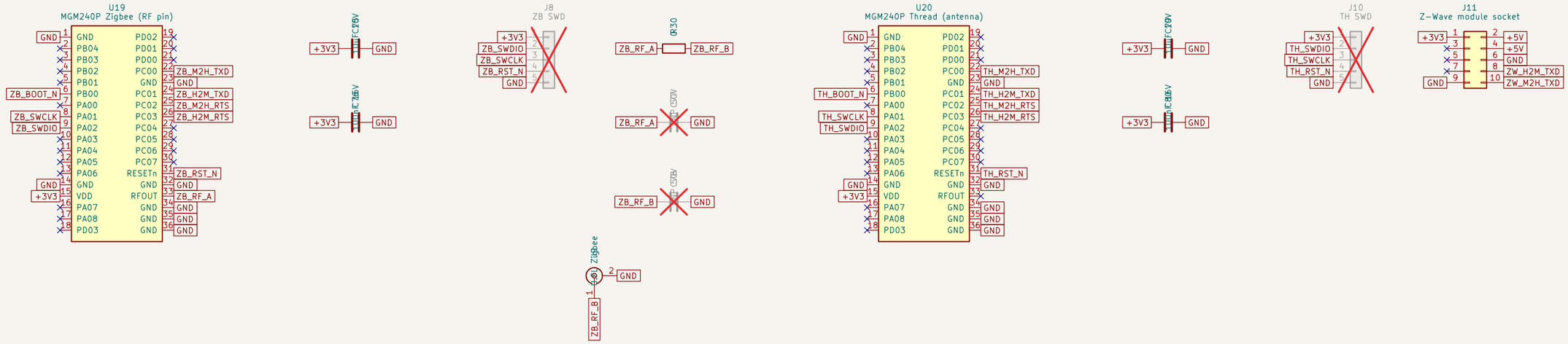
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File: ethernet_poe.kicad_sch

Title: Tahanan Hub — Gigabit Ethernet and 802.3bt PoE powered device

Size: A3	Date:	**Rev: A**
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Zigbee, Thread and Z-Wave

U19 Zigbee coordinator (EZSP NCP) with external antenna on J9. U20 OpenThread RCP with built-in antenna (Matter over Thread).
Separate radios so Zigbee and Thread never share airtime. UART with RTS/CTS, 115200 or 460800 baud. BOOT low at reset enters the
Gecko bootloader for firmware update over UART. J8/J10 are unpopulated SWD pads for factory programming and recovery.
J11 accepts a Raspberry Pi GPIO-header Z-Wave 800 module (pins 1-10 of the 40-pin header: Aeotec Z-Pi 800, Zooz ZAC93).



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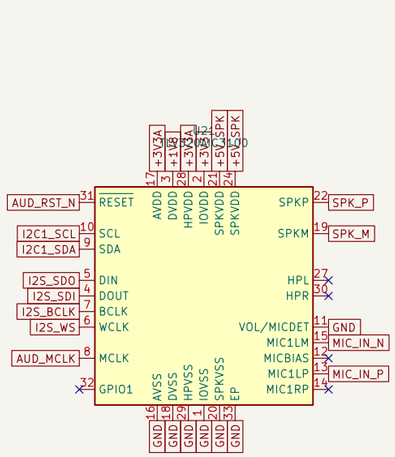
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File: radios.kicad_sch

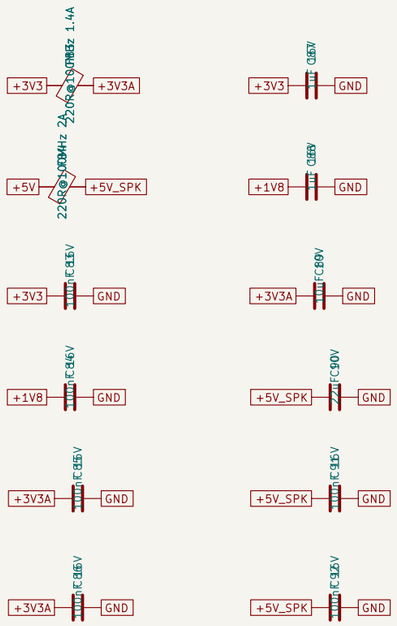
Title: Tahanan Hub — Zigbee, Thread and Z-Wave

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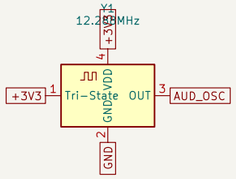
Audio codec, microphone and speaker output

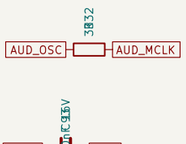
TLV320AIC3100 is the I2S clock master, clocked by Y1 (12.288MHz = 256 x 48kHz); the CM5 I2S port is the clock consumer.
One shared clock for capture and playback keeps the echo-cancellation reference sample-accurate. I2C address 0x18.
MK1 is a bottom-port analog MEMS mic mounted on the bottom side over a 1.0mm acoustic hole. SW4 removes microphone
power in hardware (privacy) and GPIO27 reports the state. Class-D output: 2.5W into 4 ohm from +5V.

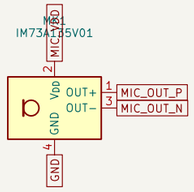


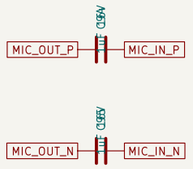


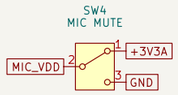




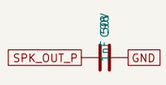


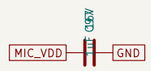




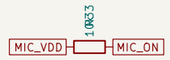






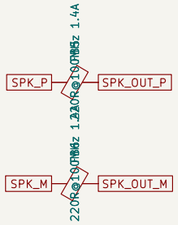


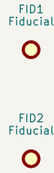












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Tahanan

Sheet: /Audio codec, microphone and speaker output/
File: audio.kicad_sch

Title: Tahanan Hub — Audio codec, microphone and speaker output

Size: A3	Date:	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 8/8